

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 3.2 PROBABILITY I – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can Kamau, and other Kenyans facing uncertain everyday outcomes, use the mathematics of probability to make smarter, fairer, and safer decisions? Kamau is a matatu conductor at Machakos Country Bus Station. His 14-seater matatu runs the Nairobi–Kitengela route. His 20-trip tally notebook records these passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14. A trip with 14 passengers is a fully booked (full) trip. Answer every section in the space under each prompt. Show your working, write fractions as well as decimals where asked, and use full sentences when you are asked to explain. Refer to Kamau's data and to what you learned in Lessons 1 to 8.

Section 1: Why can't Kamau predict one trip, yet still say something useful?

(a) Use Kamau's 20 trip counts to find the experimental probability that a trip is fully booked. Show how you counted. (b) In Lesson 1 you tossed a 10-shilling coin 30 times in pairs. No pair got exactly 15 heads, but the pooled class total was close to 0.5. Explain what this tells you about experimental probability and the number of trials. (c) In your own words, explain why Kamau cannot know for certain whether his next trip will be full, yet can still make a useful mathematical statement about his chances.

(a) Counting the trips with 14 passengers gives trips 3, 5, 7, 10, 12, 15, 17 and 20, which is 8 trips. Experimental probability = number of times the event occurs ÷ total number of trials, so $P(\text{Full}) = 8/20 = 0.4$. (b) Each pair's result was different because a small number of tosses varies a lot. When the whole class's results were pooled, there were many more trials and the fraction of heads moved closer to 0.5. This shows that more trials give a more reliable estimate of probability, even though no single toss can be predicted. (c) Each trip is a random event, so any single trip could turn out full or not full, and the notebook has no rule that predicts it perfectly. But over many trips the fraction of full trips settles near a stable value. For Kamau this is about 0.4, so he can say that roughly 2 out of every 5 trips are likely to be full. Uncertainty about one trip does not stop him measuring his chances over many trips.

Section 2: The probability scale and the sample space

(a) Kamau's matatu stops at Mlolongo and then Athi River. At each stop a particular seat is

(a) The sample space is {TT, TE, ET, EE}. I can list it using a two-way table with Stop 1 in the rows and Stop 2 in the columns, so every combination appears exactly once. That gives $2 \times 2 = 4$ outcomes, with none missing and none repeated. (b) 0.4 lies between 0 and 0.5, a little below halfway along the scale. 0.6 lies a little above halfway. They add up to 1. Neither is 0 (impossible)

either Taken (T) or Empty (E). List the complete sample space for the two stops, writing the outcomes as ordered pairs. How do you know your list is complete and has no repeats? (b) Kamau's $P(\text{Full}) = 0.4$ and $P(\text{Not Full}) = 0.6$. Place these on a 0 to 1 probability scale in words, and explain why no probability from his notebook could ever be -0.3 or 1.4. (c) Give one example of an impossible event and one certain event from Kamau's matatu world, with their probabilities.	nor 1 (certain), so a full trip is possible but not guaranteed. A probability is a count of favourable outcomes divided by the total number of trials. A count cannot be less than 0, so the probability cannot be negative. A count cannot be more than the total, so the probability cannot exceed 1. Therefore $0 \leq P(E) \leq 1$. (c) Impossible: a 14-seater matatu legally carrying 20 passengers, so $P = 0$. Certain: the matatu will eventually stop at a stage, so $P = 1$.
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Section 3: Mutually exclusive events and the Addition Law

Let F be the event 'a trip is fully booked (14 passengers)' and L be the event 'a trip has fewer than 7 passengers'. (a) Use Kamau's 20 trip counts to find $P(F)$ and $P(L)$ as fractions. (b) Are F and L mutually exclusive? Explain. (c) Find $P(F \text{ or } L)$ using the Addition Law, and say what it means for Kamau in a sentence.	(a) F: 8 trips had 14 passengers, so $P(F) = 8/20$. L: only the two trips with 6 passengers had fewer than 7, so $P(L) = 2/20$. (b) Yes. A single trip cannot be both fully booked (14 passengers) and have fewer than 7 passengers, so the two events share no outcomes and cannot happen at the same time. (c) For mutually exclusive events, $P(F \text{ or } L) = P(F) + P(L) = 8/20 + 2/20 = 10/20 = 1/2$. This means that about half of Kamau's trips are either fully booked or nearly empty. The Addition Law works here only because a trip cannot be both at once.
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Section 4: Independent events, the Multiplication Law and a tree diagram

Kamau runs a morning and an afternoon trip. Use $P(\text{Full}) = 0.4$ for each trip, and assume the two trips are independent. (a) What does it mean for the two trips to be independent? (b) Draw or describe a two-stage tree diagram with all four end outcomes and write the probability of each. (c) Find	(a) Independent means that whether the morning trip is full does not change the probability that the afternoon trip is full. Knowing the first result gives no new information about the second. (b) The first stage (morning) has branches Full 0.4 and Not Full 0.6. From each of these, the second stage (afternoon) has the same branches, Full 0.4 and Not Full 0.6. The four end outcomes are: Full–Full = $0.4 \times 0.4 = 0.16$; Full–Not Full = $0.4 \times 0.6 = 0.24$; Not Full–Full = $0.6 \times 0.4 = 0.24$; Not Full–Not Full = $0.6 \times 0.6 = 0.36$. (c) $P(\text{full on both}) = 0.4 \times 0.4 = 0.16$, which is $4/25$. $P(\text{exactly one full}) = 0.24 + 0.24 = 0.48$, because probabilities are multiplied along a branch and added across different end outcomes. (d) The four end probabilities must add to 1: $0.16 + 0.24 + 0.24 + 0.36 = 1$. This confirms the tree is correct.
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P(full on both trips) and P(full on exactly one trip). (d) Show how you can check that your tree diagram probabilities are correct.	
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Section 5: Advising Kamau – the Probability Report

Kamau's employer asks for advice. (a) Write a short Probability Report (4 to 6 sentences) saying what Kamau should expect about full buses, how he should plan, and what he cannot know. Use numbers from his notebook. (b) Suppose each of the 14 seats was filled independently with probability 10/14. The probability that all 14 seats are filled would then be about 0.009, but Kamau's data gave 0.4. What does this mismatch tell you about the independence assumption and about the model? (c) Finish this sentence in your own words: 'Probability helps Kenyans make smarter, fairer and safer decisions because...' Give one example from the unit, such as a fair game or a road-safety decision.	(a) Report: In Kamau's 20 trips, 8 were fully booked, so the experimental probability of a full trip is $8/20 = 0.4$. He should expect a full bus on about 2 out of every 5 morning departures and plan a second vehicle or extra staff for roughly that share of trips. He can budget fuel and staffing over a working week using this estimate. He cannot predict any single trip with certainty, because each trip is random and the probability of 0.4 lies between 0 and 1. More recorded trips would make his estimate more reliable. (b) The model that assumes each seat is filled independently predicts about 0.009, far below the observed 0.4. So the independence assumption must be questioned. Passengers board in clusters, for example school groups, market days and word of mouth, so seats are not filled independently. When a model's prediction contradicts the data, the data wins and the model's assumptions must be revised. Treating a full trip as a single event with probability 0.4 estimated from the notebook is more useful for planning. (c) Probability helps Kenyans because it measures uncertainty with numbers, so decisions are based on evidence rather than guesses. Example: a game is fair only if every player's theoretical probability of winning is equal, and this must be proven by calculation. Another example: the Nairobi CBD road-safety tree diagram shows how probability can measure risk and guide safer choices about where to cross.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Experimental probability and use of data	Correctly counts 8 full trips from the tally, gets $8/20 = 0.4$, and clearly links more trials to a more reliable estimate.	Gets 0.4 correctly but gives only a brief explanation of why more trials matter.	Miscounts the data or does not use the relative-frequency definition.

Sample space, probability scale and the Addition Law	Lists {TT, TE, ET, EE} correctly, explains why $0 \leq P \leq 1$, and correctly applies the Addition Law with a justification of mutual exclusivity.	Gets the sample space and the answer of 1/2 but gives a weak justification for the scale or mutual exclusivity.	Omits outcomes, gives a probability outside 0 to 1, or adds probabilities without checking the events.
Independent events, Multiplication Law and tree diagram	Defines independence accurately, completes all four branches with correct products (0.16, 0.24, 0.24, 0.36), finds 0.16 and 0.48, and uses the sum-to-1 check.	Gets most branch probabilities and the main answers right but misses the check or explains independence loosely.	Adds instead of multiplies along branches, or leaves out outcomes.
Reasoning, model critique and communication	Writes a clear report with numbers, explains why the independent-seats model fails, and links probability to fair or safe decisions in Kenyan contexts.	Gives usable advice with numbers but only a partial explanation of the model mismatch.	Gives vague advice with no numbers or no link to the data.

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